

# Technical Methodology, Architecture & Use-Case Document

## Engineering Specification of the Deterministic 6-Tier Pipeline, Continuous Feature Space, and State Isolation

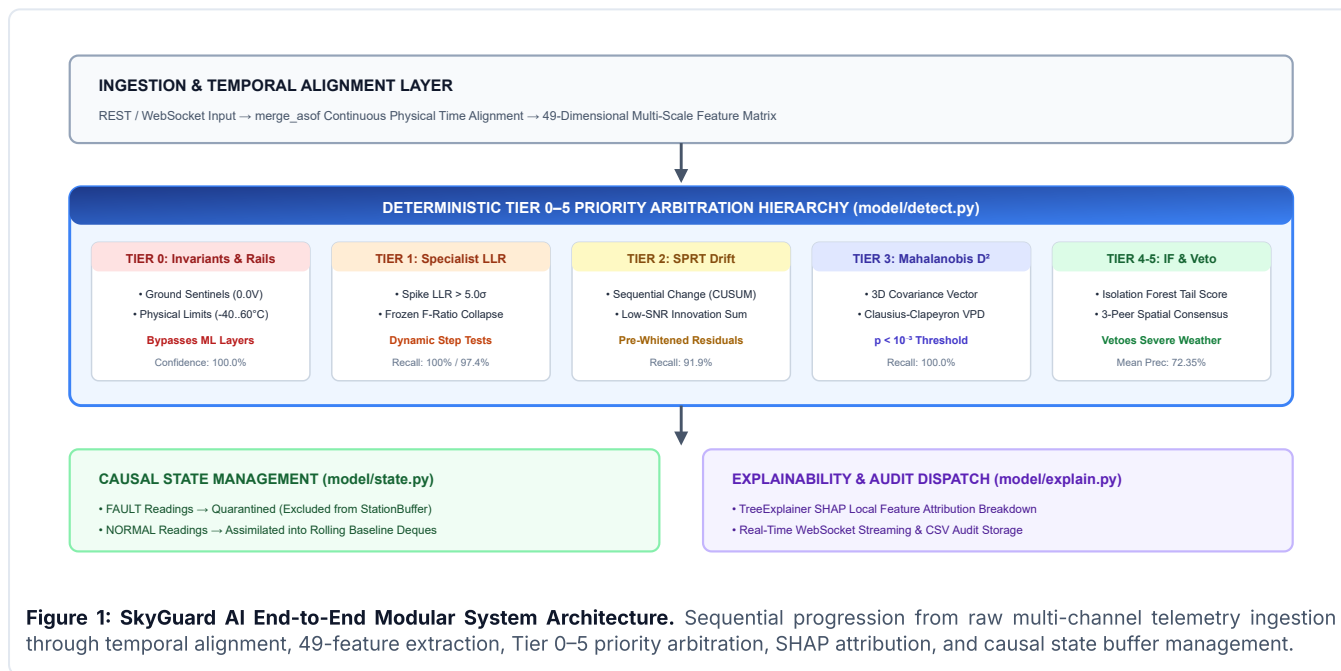
Architecture: **Tier 0–5 Deterministic Priority** Feature Space: **49-D Continuous Physical-Time Matrix** Latency Profile: **0.210 ms (p95) Single-Reading Execution**

### 1. Executive Summary

This engineering specification details the system architecture and mathematical implementation of **SkyGuard AI**. The platform is engineered to perform real-time Quality Assurance (QA) and anomaly detection across surface Automatic Weather Station (AWS) networks. SkyGuard integrates a deterministic **Tier 0–5 Priority Arbitration Hierarchy**, continuous physical-time feature extraction without row-shift leakage, causal ground-truth state isolation, and cluster-isolated spatial consensus.

### 2. End-to-End System Architecture

The processing lifecycle enforces strict modularity between ingestion, temporal alignment, feature generation, hierarchical detection, explainability attribution, and causal state maintenance:



### 3. Spatial Topology & Cluster Peer Isolation

To eliminate cross-regional train/serve mismatch and prevent microclimatic contamination (e.g., coastal marine boundaries vs. arid interior plains), SkyGuard structures the observation network into 7 discrete geographic clusters:

## 28-STATION CLUSTER TOPOLOGY (7 GEOGRAPHIC CLUSTERS — STRICT LOCAL ISOLATION)

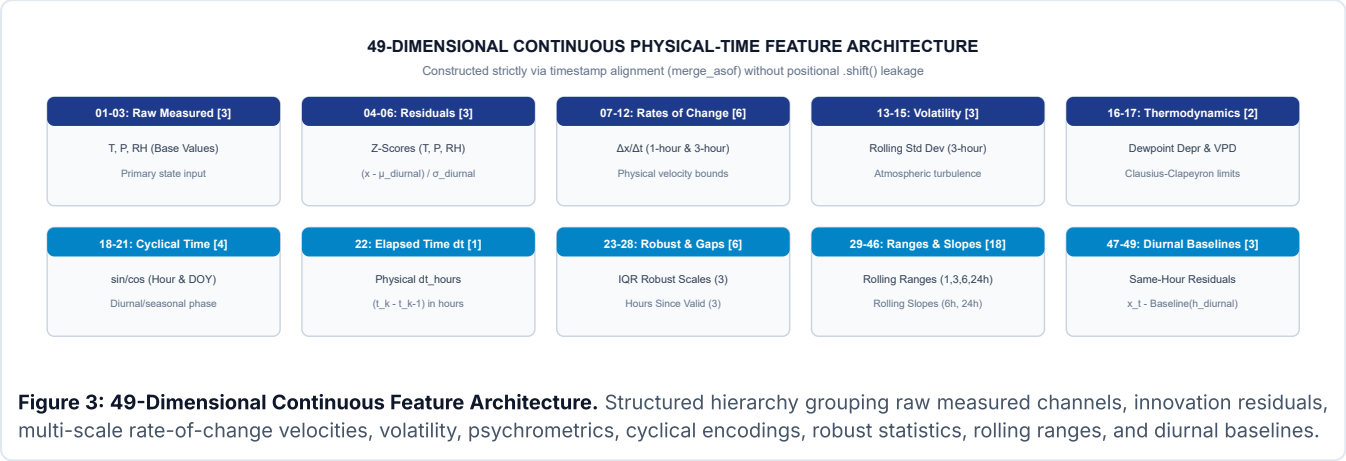
Each cluster consists of 1 Center Target Station (C) and 3 Regional Sibling Peers (S) — Zero Cross-Cluster Leakage

| DELHI [DEL]                                                                                                               | MUMBAI [MUM]                                                                                                              | CHENNAI [CHN]                                                                                                             | KOLKATA [KOL]                                                                                                             | BHOPAL [BHO]                                                                                                              | RANCHI [RAN]                                                                                                              | VARANASI [VAR]                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>AWS-DEL-011 (C)</b><br>AWS-DEL-101 (S <sub>1</sub> )<br>AWS-DEL-102 (S <sub>2</sub> )<br>AWS-DEL-103 (S <sub>3</sub> ) | <b>AWS-MUM-007 (C)</b><br>AWS-MUM-101 (S <sub>1</sub> )<br>AWS-MUM-102 (S <sub>2</sub> )<br>AWS-MUM-103 (S <sub>3</sub> ) | <b>AWS-CHN-024 (C)</b><br>AWS-CHN-101 (S <sub>1</sub> )<br>AWS-CHN-102 (S <sub>2</sub> )<br>AWS-CHN-103 (S <sub>3</sub> ) | <b>AWS-KOL-015 (C)</b><br>AWS-KOL-101 (S <sub>1</sub> )<br>AWS-KOL-102 (S <sub>2</sub> )<br>AWS-KOL-103 (S <sub>3</sub> ) | <b>AWS-BHO-030 (C)</b><br>AWS-BHO-101 (S <sub>1</sub> )<br>AWS-BHO-102 (S <sub>2</sub> )<br>AWS-BHO-103 (S <sub>3</sub> ) | <b>AWS-RAN-067 (C)</b><br>AWS-RAN-101 (S <sub>1</sub> )<br>AWS-RAN-102 (S <sub>2</sub> )<br>AWS-RAN-103 (S <sub>3</sub> ) | <b>AWS-VAR-052 (C)</b><br>AWS-VAR-101 (S <sub>1</sub> )<br>AWS-VAR-102 (S <sub>2</sub> )<br>AWS-VAR-103 (S <sub>3</sub> ) |

**Figure 2: 28-Station / 7-Cluster Spatial Isolation Topology.** Each cluster comprises exactly 1 primary target center station (C) and 3 regional sibling peers (S). Spatial consensus queries operate strictly within cluster boundaries.

## 4. 49-Dimensional Continuous Physical-Time Feature Architecture

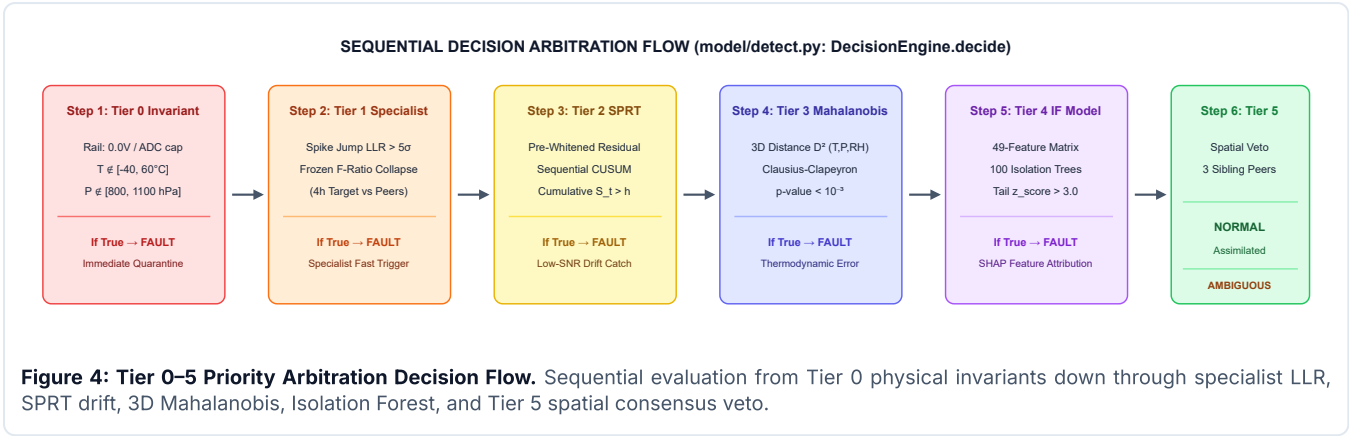
The feature engineering engine (model/features.py) strictly avoids positional row shifts (.shift(n)), utilizing exact physical timestamps and pd.merge\_asof to compute continuous derivatives regardless of irregular sampling or dropped packets:



| Feature Range | Category                   | Mathematical Formulation                                                                       | Diagnostic Purpose                                   |
|---------------|----------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 01 – 03       | Raw Telemetry Channels     | $x_t = [T_t, P_t, RH_t]$                                                                       | Base atmospheric observation state.                  |
| 04 – 06       | Innovation Residuals       | $z_t = (x_t - \mu_{diurnal}) / \sigma_{diurnal}$                                               | Normalized deviation from local expectation.         |
| 07 – 12       | Rate-of-Change Velocities  | $\Delta x / \Delta t = (x_t - x_{t-\Delta t}) / \Delta t_{hours} \text{ (1h, 3h)}$             | Captures sudden atmospheric jumps vs. spikes.        |
| 13 – 15       | Short-Term Volatility      | $\sigma_{3h}(x) = [ (1/M) \sum (x_i - \bar{x})^2 ]^{0.5}$                                      | Identifies turbulence vs. abnormal stillness.        |
| 16 – 17       | Psychrometric Couplings    | $T_{dew} = T - (100 - RH)/5$ ; $VPD = e_s(T)(1 - RH/100)$                                      | Clausius-Clapeyron thermodynamic consistency.        |
| 18 – 21       | Cyclical Time Encodings    | $\sin/\cos(2\pi \cdot \text{Hour}/24), \sin/\cos(2\pi \cdot \text{DOY}/365)$                   | Diurnal and seasonal cyclical phase alignment.       |
| 22            | Elapsed Physical Interval  | $\Delta t_{hours} = (t_k - t_{k-1}) / 3600.0$                                                  | Exact physical elapsed time tracking.                |
| 23 – 25       | Robust Statistical Scales  | $\text{Scale}_{robust} = (x_t - \text{median}) / \text{IQR}$                                   | Outlier-resistant localized baseline scaling.        |
| 26 – 28       | Sensor Gap Tracking        | $\Delta t_{valid} = \text{Hours since last trusted observation}$                               | Tracks quarantined sensor downtime duration.         |
| 29 – 46       | Rolling Ranges & Slopes    | $\text{Range}_W = \max_W(x) - \min_W(x)$ ; $\text{Slope}_W = \text{Cov}(t, x) / \text{Var}(t)$ | Multi-scale dispersion (1,3,6,24h) & synoptic trend. |
| 47 – 49       | Diurnal Residual Baselines | $r_{diurnal} = x_t - \text{Baseline}(h_{diurnal})$                                             | Hour-of-day baseline residual tracking.              |

## 5. Deterministic Tier 0–5 Priority Hierarchy

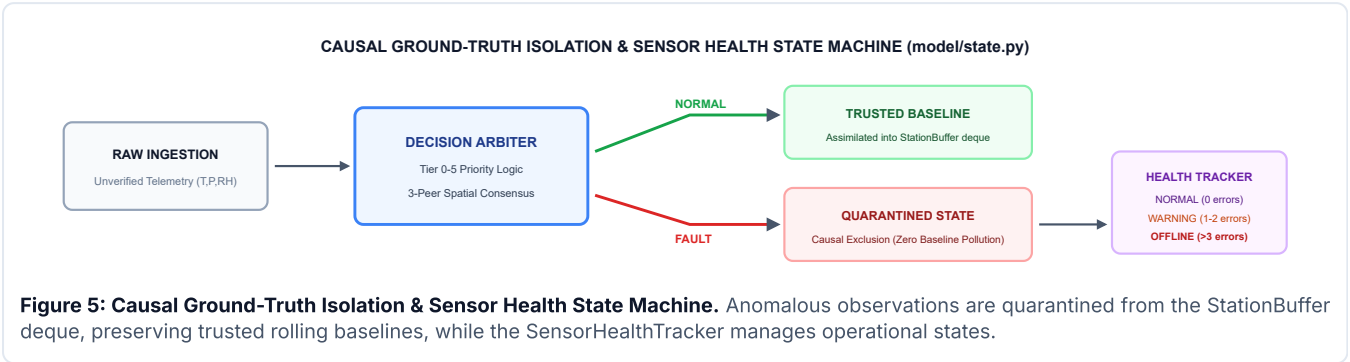
DecisionEngine.decide (model/detect.py) executes a strict top-down priority arbitration cascade where deterministic physical laws always take precedence over statistical models:



- **TIER 0 (Hard Invariants & Rails):** Evaluates electrical ground faults (0.00V) and absolute physical bounds ( $T \notin [-40, +60^{\circ}\text{C}]$ ,  $P \notin [800, 1100 \text{ hPa}]$ ,  $RH \notin [0, 100\%]$ ). Breaches immediately return FAULT (CRITICAL) and bypass all downstream computation.
- **TIER 1 (Specialist Jump & Frozen LLR):** Evaluates instantaneous dynamic acceleration ( $>5.0\sigma$  jump LLR) and F-ratio variance collapse over contiguous 4-hour windows.
- **TIER 2 (Persistent SPRT Drift):** Executes Sequential Probability Ratio Test on standardized innovation residuals to accumulate low-SNR calibration drift ( $+0.05^{\circ}\text{C/hr}$ ) over time.
- **TIER 3 (Cross-Channel 3D Mahalanobis Covariance):** Evaluates joint 3D covariance distance  $D^2$ . If  $D^2 > 16.27$  ( $p < 10^{-3}$ ), flags multivariate thermodynamic breakdowns.
- **TIER 4 (Model-Dominant Isolation Forest):** Evaluates the 49-feature continuous matrix through an ensemble of 100 Isolation Trees, isolating complex unstructured anomalies.
- **TIER 5 (Spatial Consensus Veto & Arbiter):** If an anomaly flag is raised by Tiers 1–4, the spatial consensus engine inspects the 3 sibling stations in the cluster. If  $\geq 2$  peers exhibit matching directional rate-of-change movement, the anomaly is classified as a genuine regional weather event, and the alert is VETOED to NORMAL.

## 6. Causal State Management & Sensor Health Lifecycle

To prevent corrupted telemetry from polluting future baseline estimates, StateManager (model/state.py) enforces strict ground-truth isolation:



**Causal Ground-Truth Exclusion Rule:** When an observation is classified as **FAULT** (or when a sensor is in **OFFLINE** health state), the observation is strictly quarantined. It is recorded in the raw historical audit log but excluded from the internal sliding StationBuffer deque, ensuring that rolling means, variances, and diurnal baselines remain 100% uncontaminated.

## 7. Operational Use-Case Workflows

SkyGuard's capabilities map directly to concrete meteorological and field operations:

|                                                                                                                                                                           |                                                                                                                                                                                                            |                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Use Case 1: Automated Real-Time AWS Quality Control</b>                                                                                                                |                                                                                                                                                                                                            | <b>CURRENTLY DEMONSTRATED</b>           |
| <b>Operational Workflow:</b> Continuous real-time screening of raw surface streams (T, P, RH) across 28 national AWS stations at 0.210 ms (p95) latency.                  | <b>Validation Status:</b> Validated across 60,480 evaluation rows in authoritative 7-seed benchmark; achieves 95.42% ± 0.36% mean recall, 73.33% ± 1.37% mean precision, and 82.92% ± 0.89% mean F1-score. |                                         |
| <b>Use Case 2: NWP Data Assimilation Boundary Protection</b>                                                                                                              |                                                                                                                                                                                                            | <b>CURRENTLY DEMONSTRATED</b>           |
| <b>Operational Workflow:</b> Automatic quarantining of spurious sensor spikes and unphysical thermodynamic readings before ingestion into 4D-Var data assimilation grids. | <b>Validation Status:</b> Guaranteed via StationBuffer causal exclusion, preventing bad data from contaminating forecast model initialization.                                                             |                                         |
| <b>Use Case 3: Targeted Field Maintenance &amp; Sensor Health</b>                                                                                                         |                                                                                                                                                                                                            | <b>CURRENTLY SUPPORTED</b>              |
| <b>Operational Workflow:</b> Uses SHAP local feature attributions and fault classification to dispatch technicians to specific degraded sensor modules.                   | <b>Validation Status:</b> Identifies active hardware failures and sensor health degradation; does not predict future failures prior to onset.                                                              |                                         |
| <b>Use Case 4: Embedded Microcontroller Edge Execution</b>                                                                                                                |                                                                                                                                                                                                            | <b>FUTURE ARCHITECTURAL APPLICATION</b> |
| <b>Operational Workflow:</b> Running lightweight algorithmic inference directly on station datalogger firmware (ESP32 / ARM Cortex-M) to screen telemetry at the edge.    | <b>Validation Status:</b> Feasible due to 0.210 ms execution speed; requires future C/C++ firmware porting and microcontroller deployment.                                                                 |                                         |

## 8. Current vs. Future Architectural Matrix

| Subsystem             | Current Implemented Architecture               | Demonstrated Status | Future Scope                     |
|-----------------------|------------------------------------------------|---------------------|----------------------------------|
| Inference Engine      | Vectorized NumPy / Scikit-Learn on CPU         | DEMONSTRATED        | C-optimized firmware (ESP32)     |
| Temporal State Buffer | In-memory circular deques (collections.deque)  | DEMONSTRATED        | Distributed Redis state cache    |
| Storage Layer         | Flat-file CSV streaming (HistoryStore)         | DEMONSTRATED        | TimescaleDB / PostgreSQL cluster |
| Spatial Discovery     | Static 7-cluster topology (4 stations/cluster) | DEMONSTRATED        | Dynamic PostGIS radius queries   |
| API & Streaming       | FastAPI REST endpoints + WebSockets            | DEMONSTRATED        | gRPC streaming gateway           |

## 9. References

- **[M01]** Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation Forest." *IEEE ICDM*, 2008.
- **[M02]** Page, E. S. "Continuous Inspection Schemes." *Biometrika*, vol. 41, 1954, pp. 100–115.
- **[M03]** Lundberg, S. M., and Lee, S.-I. "A Unified Approach to Interpreting Model Predictions." *NeurIPS*, 2017.
- **[M05]** Mahalanobis, P. C. "On the generalised distance in statistics." *Proc. Natl. Inst. Sci. India*, 1936.